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PHYSIOLOGICAL RESPONSES AND ACTIVITY PROFILES OF FOOTBALL SMALL-SIDED GAMES

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ABSTRACT

Aguiar, MVD, Botelho, GMA, Gonçalves, BSV, and Sampaio, JE. Physiological responses and activity profiles of football small-sided games. *J Strength Cond Res* 27(5): 1287–1294, 2013—The aim of this study was to identify the acute physiological responses and activity profiles of football small-sided games (SSG) formats. Ten professional football players participated in 4 variations of SSG (2-, 3-, 4-, and 5-a-side) with an intermittent regime involving 3 × 6-minute bouts with 1 minute of passive planned rest in which the heart rate (HR), rating of perceived exertion (RPE), activity profile, and body load were recorded. The higher percentage of maximum HR values were found in 2- and 3-a-side formats ($p \leq 0.05$). The lowest RPE value was found at the 5-a-side, and the highest was found at the 2-a-side (13.48 ± 2.67 and 17.01 ± 1.80 , respectively, $p \leq 0.05$). The distance covered in the 2-a-side format (598.97 ± 78.91 m) was smaller than in all other formats. The 2-a-side format presented the lowest number of sprints (0.71 ± 0.86) and the 3-a-side the highest (2.50 ± 1.65). Statistically significant differences were found across SSG in the total body load. The 4-a-side presented the highest and the 5-a-side the lowest values (95.18 ± 17.54 and 86.43 ± 14.47 , respectively). The body load per minute declined each 2 minutes of play. Maintaining a constant area:player ratio, coaches can use lower number of players (2- and 3-a-side) to increase cardiovascular effects but use higher number of players (4- and 5-a-side) to increase variability and specificity according to the competition demands.

KEY WORDS team sports, conditioning, activity profile analysis, body load, perceived exertion

INTRODUCTION

Small-sided games (SSG) are widely used by football coaches at all levels of the game to develop technical and tactical abilities and physical conditioning of players (17,33). Recent research suggests that

physiological, perceptual (e.g., heart rate [HR], blood lactate, and perceived exertion), and technical responses can be modified during SSG in football by modifying game constraints such as the number of players, the size of the pitch, the rules of the game, or coach encouragement (8,11,13,16–20,24,28,33). In general, these studies have shown that SSG formats with fewer players elicit higher HR (30) values than SSG formats with higher number of players (18,19,24–26,33,36). However, some authors have reported no differences between SSG formats (13,17,21,36).

Currently, the use of Global Positioning System (GPS) technology allows a faster gathering of time-motion data and in higher quantities, contributing to the increase of information regarding SSG responses. The validity and reliability of commercially available GPS receivers have already been established and described (9,12,14,22,27,32), allowing the increase of the accuracy of game-related conditioning activities (5,10,29). The GPS devices have an acceptable validity and reliability for recording movement patterns at lower speeds and higher sample rates over longer efforts. However, there is a lower accuracy when assessing movement during rapid variations in speed over short distances (22).

Research on time motion analysis of SSG using GPS devices is very recent. One study is available and examines the acute physiological and perceptual responses and the activity profile during SSG in three formats (2-, 4-, and 6-a-side) involving youth football players (19). The authors used a constant ratio of player number to pitch area in the different SSG formats. The results showed increases in physiological and perceptual workload as SSG formats decrease in size while maintaining the relative pitch area constant. Another study was performed with elite female players and focused on describing the activity profiles of 2 different SSG formats (3- and 5-a-side) in comparison with competition (16). The authors concluded that SSG simulate the overall movement patterns of women's football competition but offered an insufficient training stimulus to simulate the high-intensity repeated-sprint demands of competition.

The use of some commercially available GPS units with incorporated triaxial accelerometers allows the measurement of the body load associated with game play (3). Montgomery et al. (31) used these devices to study the physical demands

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of basketball training and competition and showed that these data, in combination with HR, can be useful to differentiate the physical and physiological demands during basketball practice and competition. In football, Casamichana et al. (7) identified a body load per minute of 15.8 ± 2.7 when playing 3-, 5-, and 7-a side SSG formats on different pitch layouts (without goals, with 2 regulation goals and goalkeepers, and with 2 small goals, but no goalkeepers). The authors presented this as the overall result and did not compare the different SSG formats. Also, an interesting topic to add to this study would be to compare body load per minute across the duration of play to identify different patterns associated with different SSG.

A recent literature review (10) has identified that very few studies have examined the influence of the concurrent manipulation of the pitch area and SSG intensity of player numbers. Additionally, other important aspects were not considered in the previous studies, such as the influence of pitch size and player numbers in the relative pitch area, and these 2 factors were regularly changed throughout the studies (6,7,17,23,31–33). Only one study is available with a constant area:player ratio focusing only on 2-, 4-, and 6-a side SSG formats in players under 17 years (19). Therefore, the aim of this study was to examine the acute physiological responses, activity profile, and body load of 4 football SSG formats (2-, 3-, 4-, and 5-a-side) while maintaining the pitch size per player constant.

METHODS

Experimental Approach to the Problem

Few studies have examined the HR responses, rating of perceived exertion (RPE), activity profile characteristics, and body load of 4 different SSG formats keeping the area per player constant (10). In this study, four SSG formats were used (2-, 3-, 4- and 5-a-side players) with an intermittent regime involving 3×6 -minute bouts with 1 minute of passive planned rest (work:rest ratio = 6:1). This protocol was repeated twice. The relative pitch size per player remained constant across the different game formats at around 150 m² per player.

Subjects

Ten professional male football players younger than 19 years of age (age = 18.0 ± 0.67 years, body mass = 69.7 ± 5.66 kg, height = 1.79 ± 0.06 m, body mass index = 21.83 ± 0.98 kg·m⁻², maximum heart rate [HR_{max}] = 193.9 ± 5.0 ; mean $\pm$ SD) participated in this study. All players were members of the same youth team (<19 years) competing in the Portuguese First Division. The team had practice sessions 5 times a week, 90 minutes per session, with an official game during the weekend. Players had a mean of 10.20 ± 1.81 years of experience (minimum of 7 years and maximum of 12 years) and 5 of these players had already represented the National team. Furthermore, all players practiced regularly with the senior professional team that competed in the Portuguese 1st League.

All players and their parents were informed about the research procedures, requirements, benefits, and risks, and their written consent was obtained before the study began. The study was approved by the Ethics Committee of the Research Center for Sports Sciences, Health and Human Development (Vila Real, Portugal) and by the local ethics committee before the commencement of the assessments. The study protocol followed the guidelines described in the Declaration of Helsinki.

Teams' Constitution

The games were played in the last 7 weeks of the competitive season. The coach used subjective evaluation to select the best 10 players, and, afterward, the players were assigned to 2 teams using specific positions, tactical/technical levels, physical performance, and participation in competitive matches (8). The players' tactical/technical level was established according to the subjective evaluation carried out by the coach, who awarded a score of 1 to players with the lowest level and 5 to those with the highest level. The number of minutes played in competitive matches (before the beginning of this study) was also used to categorize players; a score of 1 was given to players with the fewest number of minutes played and 5 to those who played the most minutes. Scores on the Yo-Yo intermittent recovery test level 2 (1) were used in the same way: players who covered the least distance were given a score of 1 and those who performed the greatest distance were classified with 5. The total score of each player was the sum of the 3 scores.

Procedures

Before the first data collection session, a theoretical presentation was given to present and discuss the procedures with the participants. Also, an initial session was conducted with an SSG involving all 10 participants to familiarize them with the procedures and equipment for data collection. The Yo-Yo intermittent recovery test level 2 (1) was also conducted in this session.

Each SSG format was played twice by the same teams in different sessions. The SSGs were played at the beginning of each training session, after a standardized 20-minute warm-up, consisting of running, stretching, and a ball possession game. Players played an SSG with 20-minutes duration (3×6 -minute bouts with 1-minute of active recovery). After each period of 6 minutes, teams changed the attack direction. During the SSG games, formal goals were used and a goalkeeper was added randomly to each team. When the ball was kicked out of play, immediate access to a replacement football was made possible by having a supply of balls placed in the goals and along the boundary line surrounding the entire pitch. Unlimited touches on the ball were allowed, and offside rule was not applied during the game. Furthermore, coach encouragement or feedback was not allowed during SSGs.

Field dimensions used during different SSG formats were manipulated to keep the relative area per athlete constant (19). The ratio was kept in 1:150 m² per player.

Rating of Perceived Exertion

Global RPE were recorded immediately after each SSG using the 6–20 Borg scale (2). Standardized instructions for RPE were provided. Players were already familiarized with the 6–20 Borg scale before this study.

Heart Rate Monitoring

The HR of each player was recorded at 5-Hz intervals during each SSG via short-range radio telemetry (Polar Team Sports System; Polar Electro Oy, Kempele, Finland). The HR monitors were also used during the Yo-Yo intermittent recovery test level 2 (1), to determine each player's HR_{max}. Exercise intensity during each SSG was assessed using HR, expressed as a percentage of maximum heart rate (%HR_{max}) and classified into 4 previously defined zones: zone 1 (<75% HR_{max}), zone 2 (75–84% HR_{max}), zone 3 (85–90% HR_{max}), and Zone 4 (>90% HR_{max}) (19). The HR_{max} reached during Yo-Yo intermittent recovery test level 2 (1) was used as the reference (100%) value.

Activity Profile

Player movements during the SSG were measured using portable GPS units, sampled at 5 Hz (SPI-PRO; GPSports, Canberra, Australia). This equipment operated in nondifferential mode and provided data in real time. The GPS is a satellite-based navigation system that enables real-time data during training and competition (9,12,27,29). The reliability of the SPI-PRO GPS device for measuring specific movement characteristics in team sports after a known running course was previously examined (9,14,27). The authors found that the GPS devices have an acceptable level of accuracy and reliability for total distance and peak speeds during high-intensity and intermittent exercise. The SPI-PRO was placed into a harness that positioned the device between the player's shoulder blades, which was used by each player during all SSG (19). For data analysis purposes, six speed zones were selected (17): speed zone 1 (0–6.9 km·h⁻¹), speed zone 2 (7.0–9.9 km·h⁻¹), speed zone 3 (10.0–12.9 km·h⁻¹), speed zone 4 (13.0–15.9 km·h⁻¹), speed zone 5 (16.0–17.9 km·h⁻¹), and speed zone 6 (sprinting: ≥18.0 km·h⁻¹).

Body Load

Body load is a modified vector magnitude (3), based on the accelerations detected by the accelerometer from 3 planes (vertical, horizontal, and anterior-posterior) at 100 Hz. The registered values are internally stored and converted into quantifiable digital signals (39). This variable is expressed, as showed in the following equation, as the square root of the sum of the squared instantaneous rate of change in acceleration in each of the three vectors (x, y, and z) and divided by 100. Data were expressed in arbitrary units. The body load validity has already been established by Boyd et al. (3).

$$\text{Body load} = \frac{\sqrt{(a_{y1}-a_{y-1})^2 + (a_{x1}-a_{x-1})^2 + (a_{z1}-a_{z-1})^2}}{100}$$

where a_y = forward accelerometer, a_x = sideways accelerometer, and a_z = vertical accelerometer. The body load results were measured and compared for each minute of the SSG play.

Statistical Analyses

The data are reported as mean ± SD. The HR, %HR_{max}, RPE, distance traveled, and total body load were compared across the SSG formats using a 1-way analysis of variance (ANOVA). The Bonferroni post hoc test was used to identify pairwise comparisons. The body load, total distance, and distances covered above and below 16 km·h⁻¹ were also analyzed per minute by using a repeated-measures ANOVA 6 × 4 (minute: 1, 2, 3, 4, 5, and 6 and SSG format: 2-, 3-, 4-, and 5-a-side). These results were presented with curves fitted according to the distance-weighted least squares smoothing procedure, that is, the influence of individual cases decreases with the horizontal distance from the respective cases on the curve. The Bonferroni post hoc test was used to identify pairwise comparisons. To compare the HR zones and speed zones between SSG formats, the nonparametric Wilcoxon Signed-Rank Test was used with the Bonferroni adjustment. The level of statistical significance was set at $p \leq 0.05$. All statistical analyses were performed using the software IBM SPSS statistics (version 19; SPSS, Inc., Chicago, IL, USA).

RESULTS

Physiological and Perceptual Characteristics

Table 1 presents the time spent in each HR zone during the various SSG formats. One-way ANOVA identified statistically significant differences among SSG formats in the %HR_{max} ($F = 5.93$; $p = 0.001$). Statistical differences between SSG formats were also found in all times spent in HR zones, except time spent at 75–84% HR_{max}.

The lowest RPE value was found at the 5-a-side format (13.48 ± 2.67) and the highest value (17.01 ± 2.88) was found at the 3-a-side format. In the 2- and 4-a-side formats, the RPE values were 17.01 ± 2.88 and 15.00 ± 2.25 , respectively. The 1-way ANOVA showed significant differences ($F = 6.08$; $p = 0.001$) between RPE values in the SSG formats. The Bonferroni post hoc test showed significant differences between the RPE values of 2- vs. 5-a-side formats and 3- vs. 5-a-side formats.

Activity Profile

Table 2 presents the total distance covered in each SSG and the distance covered in the 6 speed zones. Statistically significant differences among SSG formats were found for total distance performed ($F = 7.24$; $p = 0.000$). The distance covered in the 2-a-side format was statistically different from all other formats. In the 2-a-side, the players performed the smallest distance in each 6-minute period (598.97 ± 78.91 m), whereas the largest distance was covered in the 3-a-side format (685.71 ± 72.77 m). Considering the time spent in each speed zone, only zone 5 and 6 did not present statistically significant differences between SSG formats.

TABLE 1. Percentage of heart rate in the different SSG formats (mean $\pm$ SD).

SSG format	2-a-side	3-a-side	4-a-side	5-a-side
%HRmax	87.46 $\pm$ 7.46	89.56 $\pm$ 3.15 ^{d,e}	85.91 $\pm$ 5.98	84.56 $\pm$ 7.56
Time at <75% HRmax (31)	0.44 $\pm$ 0.50 ^a	0.30 $\pm$ 0.21 ^{d,e}	0.65 $\pm$ 0.84	1.02 $\pm$ 1.25
Time at 75–84% HRmax (31)	1.05 $\pm$ 0.91	0.68 $\pm$ 0.61	1.52 $\pm$ 1.09	1.64 $\pm$ 1.25
Time at 85–89% HRmax (31)	1.61 $\pm$ 1.02 ^a	1.52 $\pm$ 1.11 ^{d,e}	1.67 $\pm$ 1.05 ^f	1.27 $\pm$ 1.17
Time at >90% HRmax (31)	2.91 $\pm$ 1.42 ^a	3.50 $\pm$ 1.44 ^e	2.14 $\pm$ 1.78 ^f	2.07 $\pm$ 1.97

Statistical significant differences are identified as: ^a2-a-side vs. 3-a-side; ^d3-a-side vs. 4-a-side; ^e3-a-side vs. 5-a-side; ^f4-a-side vs. 5-a-side. SSG = small-sided game; %HRmax = percentage of maximum heart rate.

The number of sprints (distance covered with a speed ≥ 18 km $\cdot$ h⁻¹), performed by players during the 4 SSG can also be found in Table 2. Statistically significant differences were found between the number of sprints performed in each SSG ($F = 7.82$; $p = 0.000$). The 2- and 3-a-side formats present differences when compared with all other formats. The highest number of sprints was performed in the 3-a-side format (2.50 ± 1.65) and the lowest in the 2-a-side format (0.71 ± 0.86).

Total body load and body load per minute in each SSG format are presented in Table 2. Statistically significant differences among the SSG formats were found in both variables, ($F = 4.03$, $p = 0.008$ and $F = 3.83$, $p = 0.010$, respectively). Statistically significant differences between 4- and 5-a-side SSG were found. The 5-a-side format presented the lowest total body load (86.43 ± 14.47), whereas the 4-a-side presented the highest value (95.18 ± 17.54). The same tendency was found in body load per minute, with 14.42 ± 2.41 and 15.88 ± 2.93 , respectively. Figure 1A presents the body load per minute in each SSG. There was a single effect of minute

of play ($F = 20.27$; $p = 0.000$) with lower values at the end of each SSG. There were differences in all pairs of minutes with the exception of the following: minute 1 and 2, minute 3 and 4, and minute 5 and 6. There was also an interaction with SSG formats ($F = 1.73$; $p = 0.040$). There were differences in the following pairs: 2- and 4-a-side, 3- and 5-a-side, and 4- and 5-a-side (Figure 1A). Figure 1B presents the total distance covered per minute with a significant effect of minute of play ($F = 10.8$; $p = 0.000$) and a significant interaction with SSG formats ($F = 3.19$; $p = 0.000$). The total distance covered was significantly lower in minutes 5 and 6. Also, the 2-a-side game was different from all other formats. These results were similar for the distance covered per minute at lower speeds ($F = 9.0$; $p = 0.000$ for minute of play and $F = 2.74$; $p = 0.000$ for interaction with SSG formats). Finally, the results for distance covered per minute at higher speeds were not significant either for the effect of minute of play ($F = 1.8$; $p = 0.098$) and for the interaction with SSG formats ($F = 1.73$; $p = 0.161$, Figure 1D).

TABLE 2. Activity profile characteristics (mean $\pm$ SD) regarding 6 speed zones (m), total distance covered (m), sprints/athlete, and body load during all SSG formats.

SSG format	2-a-side	3-a-side	4-a-side	5-a-side
Zone 1 (0–6.9 km $\cdot$ h ⁻¹)	291.84 $\pm$ 18.99 ^{a,b,c}	278.44 $\pm$ 26.50	272.74 $\pm$ 30.62	285.30 $\pm$ 26.82
Zone 2 (7.0–9.9 km $\cdot$ h ⁻¹)	137.41 $\pm$ 34.16 ^b	141.14 $\pm$ 33.81 ^d	163.73 $\pm$ 40.36	143.10 $\pm$ 37.81
Zone 3 (10.0–12.9 km $\cdot$ h ⁻¹)	95.19 $\pm$ 38.05 ^{a,b,c}	126.30 $\pm$ 38.12	128.38 $\pm$ 40.06	117.09 $\pm$ 39.31
Zone 4 (13.0–15.9 km $\cdot$ h ⁻¹)	48.46 $\pm$ 24.68 ^{a,b,c}	79.51 $\pm$ 22.37	72.59 $\pm$ 24.00	67.40 $\pm$ 28.47
Zone 5 (16.0–17.9 km $\cdot$ h ⁻¹)	15.58 $\pm$ 9.14 ^{a,b}	30.91 $\pm$ 11.41	23.61 $\pm$ 11.97	25.09 $\pm$ 15.23
Zone 6 (≥ 18.0 km $\cdot$ h ⁻¹)	10.48 $\pm$ 9.92 ^{a,b}	29.42 $\pm$ 15.14 ^e	21.09 $\pm$ 17.23	21.99 $\pm$ 16.96
Total distance	598.97 $\pm$ 78.91 ^{a,b,c}	685.71 $\pm$ 72.77	682.14 $\pm$ 76.56	659.98 $\pm$ 81.78
Sprints/Athlete	0.71 $\pm$ 0.86 ^{a,b,c}	2.50 $\pm$ 1.65 ^{d,e}	1.71 $\pm$ 1.43	1.74 $\pm$ 1.41
Acc load (a.u.)	88.63 $\pm$ 20.37	92.32 $\pm$ 12.81	95.18 $\pm$ 17.54 ^f	86.43 $\pm$ 14.47
Acc load per min (a.u.)	14.74 $\pm$ 3.37	15.33 $\pm$ 2.14	15.88 $\pm$ 2.93 ^f	14.42 $\pm$ 2.41

Statistical significant differences are identified as: ^a2-a-side vs. 3-a-side; ^b2-a-side vs. 4-a-side; ^c2-a-side vs. 5-a-side; ^d3-a-side vs. 4-a-side; ^e3-a-side vs. 5-a-side; ^f4-a-side vs. 5-a-side. SSG = small-sided game; Acc = accumulated; a.u. = arbitrary units.

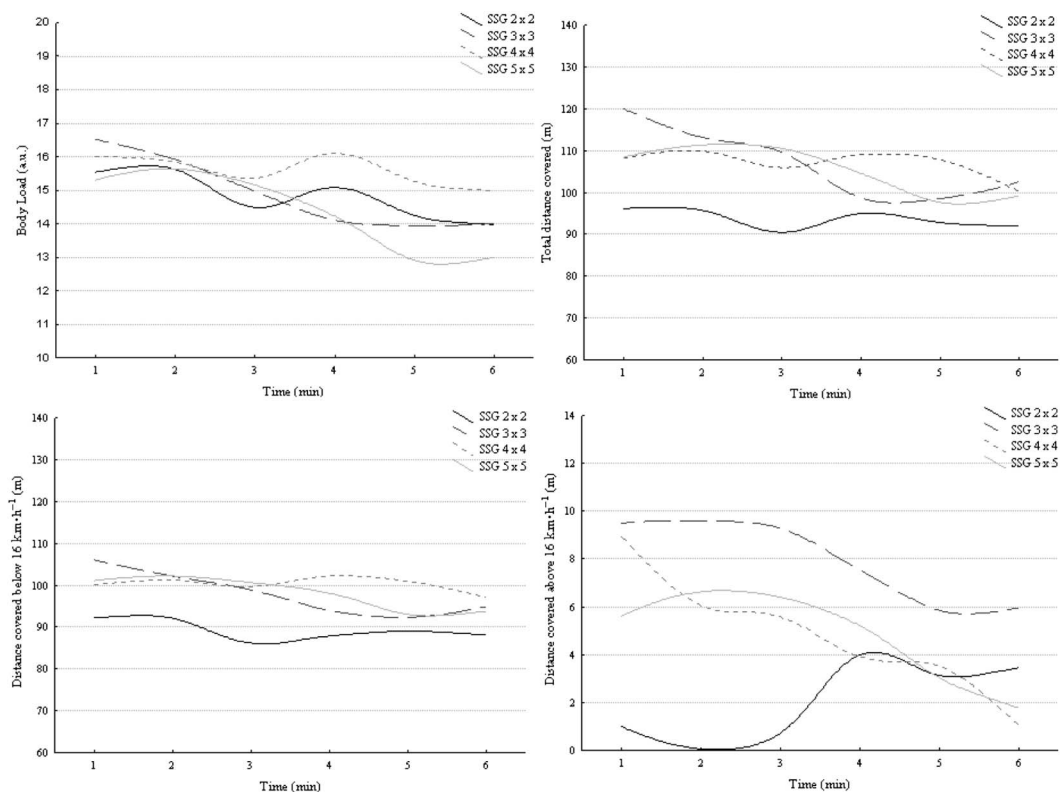


Figure 1. Body load, total distance covered, distance covered below 16 km·h⁻¹·min⁻¹, and distance covered above 16 km·h⁻¹·min⁻¹ during the 4 SSGs. SSG = small-sided game.

DISCUSSION

The aim of this study was to examine the acute physiological responses, activity profile, and body load of 4 football SSG formats (2-, 3-, 4-, and 5-a-side) while maintaining the pitch size per player constant. Previous research identified 2 important aspects not considered in earlier SSG football studies, such as changing the pitch size and the number of players, resulting in several differences in the pitch area per player (10,13,23,24,33). In this study, the area: number of players ratio was constant across all SSG formats, with the aim of characterizing HR, perceptual responses, activity profile, and body load.

Significant differences were identified in all studied variables. When differences were found, these occurred mainly among the 2- and 3-a-side formats with all the other formats. Monitoring the HR response of players during training is a useful and reliable method for regulating exercise intensity (21,35). In this study, the exercise intensity was determined by the HR and %HR_{max}. The results demonstrated that HR values obtained in 3-a-side games were significantly higher than 4- and 5-a-side games. The HR in the 2-a-side game was also significantly higher than in the

5-a-side format. Despite the fact that statistically significant differences between 2- vs. 4-a-side and between 4- vs. 5-a-side games were not found, results, in general, agree with those obtained by several authors (19,23,26,33).

In this study, the 2- and 3-a-side formats elicited a greater amount of time spent at $\geq 90\%$ HR_{max} than the 4- and 5-a-side games. Additionally, the HR responses during 5-a-side games were below the mean HR reported during a competitive match (35). Moreover, the players spent less time under previously reported match intensity zones (35) and more time over match intensity in the 2- and 3-a-side format compared with the 4- and 5-a-side formats. The results demonstrated that smaller game formats elicit higher mean HR and that players spent more time in higher %HR zones. One possible explanation for these results is that an increasing number of players might cause an insufficient involvement, with or without the ball, in the game. Moreover, the presented results suggest that these smaller game formats may be useful in training to improve aerobic fitness in football players because they can elicit HR responses around 90% of HR_{max} (21).

Available research suggests that RPE is a valid indicator of high-intensity intermittent exercise, when compared with

HR and blood lactate concentration during football-specific exercise (11). In this study, 2- and 3-a-side formats elicited a statistically significant greater RPE value than the other 2 formats (4- and 5-a-side). The last format (5-a-side) presented the lowest value (15.00 ± 2.25), whereas the 2- and 3-a-side formats presented the highest RPE values (17.01 ± 2.88 and 16.83 ± 1.80 , respectively). These results are similar to those previously reported (19,33,36) and suggest that RPE increases when the number of players decline. In accordance with the current research, it appears that an increase in the number of players with constant pitch area per player reduces the player's rating of perceived exertion in football SSG. Another explanation for the reduction in RPE with the increasing number of players may be the decreasing interaction with colleagues and opponents (19).

The analysis of the activity profile allows the exercise intensity to approach competition demands (6). The activity profile results showed in this study confirmed the results obtained in the physiological parameters. The total distance performed by players was similar across the 3 larger formats and significantly smaller in the 2-a-side format. Probably, the smaller distances performed in the 2-a-side format reflect less absolute pitch space to cover during the game.

The highest distance (291.84 ± 18.99 m) performed at low intensity (zone 1 $< 7 \text{ km} \cdot \text{h}^{-1}$) was found in the 2-a-side format, and the smallest distance was found in the 4-a-side format (272.74 ± 26.82 m). These results may be explained by the highest frequency of individual actions during the 2-a-side format and the lowest absolute space to cover during the 2-a-side game.

In general, the larger game formats were associated with greater ranges of distances traveled at speeds $18 \text{ km} \cdot \text{h}^{-1}$ or more (10,19). However, in this study, the 3-a-side format was characterized by significantly longer distances for speeds $18 \text{ km} \cdot \text{h}^{-1}$ or more. The number of sprints (displacements with speed $\geq 18 \text{ km} \cdot \text{h}^{-1}$) performed across the SSG formats was significantly lower in the 2-a-side and higher in the 3-a-side format. In the 4- and 5-a-side formats, the number of sprints was similar.

The work rates are influenced by the positional role of players and style of play (23,34). In fact, previous research on football has identified differences between playing formations (4:4:2, 4:3:3, and 4:5:1) particularly for very high-intensity running activity and some technical elements of performance. These differences were particularly noted in attackers (4). In the smaller formats, all players are required to perform defensive and offensive actions. Indeed, the present results indicate that SSG offered a specific training stimulus to simulate the overall movement patterns in competition. However, these training exercises did not simulate the high-intensity demands of the competition because all SSG formats elicit a maximum of approximately 7% of displacements with high intensity ($> 16 \text{ km} \cdot \text{h}^{-1}$) compared with a value of 10% referred to by other authors (34,35). In agreement with previous literature (19), it is possible that the activity profiles during SSGs can

be determined by the complex interaction of total pitch area, number of players, opportunity for direct involvement with the ball, and match style. In fact, very recent literature suggests that sprints should be defined as any movement that reaches or exceeds the sprint threshold velocity for at least 1 second and any movement with an acceleration that occurs within the highest 5% of accelerations found in the corresponding velocity range (15). Using this new definition will probably lead to an increase in the number of sprints recorded during SSG.

The football players' game workload is intermittent, maximal, and aerobic based, with 1,000–1,350 changes in activity during the 90 minutes, which means a change of activity every 4–6 seconds (30). The recorded accelerations or decelerations give a description of the frequency, time, and intensity of the physical activity produced by body movement (37). The accumulated body load increased with the number of players until the 4-a-side; however, in the 5-a-side results, a significant decrease to lower values than 2-a-side format was verified. A possible explanation for the highest body load found in the 4-a-side game could be the complex interactions with a wider pitch area, 3 teammates and 4 opponent players, and still having very frequent opportunities for direct involvement with the ball. Moreover, the 4-a-side is the format with the lower number of players where all defensive (delay, concentration, balance, and control/restraint) and offensive (penetration, width, mobility, and improvisation) principles of the game can be found (38). On the other hand, the 5-a-side game is the first SSG to present a game organization where the players are required to assume specific playing positions, otherwise the team will not be able to cover all the pitch adequately. Therefore, this playing strategy implies a decrease in direct involvement with the ball and, consequently, a decrease in body load.

The accumulated load per minute values are very similar to those obtained by Casamichana et al. (7) during SSG (15.8 ± 2.7) and were higher to those obtained in friendly matches (13.5 ± 1.5). These results suggest that SSG are played at a higher intensity than friendly matches. Contrarily, Montgomery et al. (31) found results that suggest that body load was greater during basketball game play than during practice drills. The same authors found body load per minute values in defensive and offensive drills similar to those we found in the SSG formats. One interesting novelty in this study was the fact that body load values decreased each 2 minutes of game play along with decreases in total distance and distance below $16 \text{ km} \cdot \text{h}^{-1}$ only after 4 minutes of play. Nevertheless, no statistically differences were found in the distance performed above $16 \text{ km} \cdot \text{h}^{-1}$, probably because of the variability and unpredictability of the game. Therefore, these results suggest that after 2 minutes of play (0–2 minutes), the accelerations/decelerations are impaired, however, with no consequences yet for distance covered at lower speeds. In the following 2 minutes (2–4 minutes), there was additional impairment in accelerations/decelerations but now with less distance covered at lower speeds. The last 2 minutes (4–6 minutes) showed lower accelerations/decelerations with no consequences for

distance covered at lower speeds. These results may suggest that SSG duration can be programmed in 2-minute blocks interspersed with very short periods of rest, to allow for a quick recovery and maintenance of the workload characteristics. These concerns are particularly important when using 5-a-side games because at the end of the game period, the body load has substantially decreased.

PRACTICAL APPLICATIONS

The results of our study showed that physiological and perceptual responses are higher in the smaller SSG formats. Additionally, results suggest that smaller game formats may be useful for training to improve aerobic fitness in football players because they can elicit HR responses around 90% of HRmax. This could indicate that smaller formats are more appropriate to increase physiological stress, whereas larger formats can be used to improve match-specific demands. This information is useful for coaches because they can modify or introduce rules in the SSG formats to adjust them to the competition demands. The body load was influenced by the SSG with the results suggesting that coaches can manipulate the duration of play in 2-minute periods. While maintaining a constant area:player ratio, coaches can use lower number of players (2- and 3-a-side) to increase cardiovascular demands but higher number of players (4- and 5-a-side) to increase variability and specificity. Activity profile and body load were not as different as expected; however, 4-a-side SSG presented higher values and higher variability and, again, the 3-a-side SSG was the most stable format.

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